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**Abstract**

**Motivation & Background:** Can humans influence the manifestation of reality through the power of their minds alone? I would like to address this question, building on the promising results of my study from an empirical internship supervised by Prof. Dr. Markus Maier.

**Research Question:** Can an incentive given to participants to reveal the red card from among two face-down cards (red and black) increase the probability of the desired red card appearing?

**Methods:** The experiment was conducted online; participants were randomly assigned to either the experimental group (given an incentive) or the control group (no incentive given). The first round of data collection included 283 participants. I supplemented the dataset in the second round with an additional 330 participants, bringing the total sample size to 613 participants.

**Results:** In the first survey, more red cards were revealed in the experimental group (with an incentive), but neither a Bayesian t-test for independent samples nor a chi-square test yielded statistical significance. In the second survey, more black cards were revealed in the experimental group, again without statistical significance. In the combined analysis of the datasets, the experimental group showed a random distribution of red and black cards. The control group showed a random distribution of cards at each assessment time point.

**Implications:** The results support existing assumptions about the inconsistent nature of micro-PK effects and provide interesting avenues for future research in this area.

*Keywords:* Micro-PK, psi research, mind-matter interaction, random biases

## Introduction

It is human nature to seek to understand the universe. This need is not limited to the discovery of alien planets and distant galaxies, but also encompasses a desire to understand one's own position and role in the universe. Humans wish to know how their actions and decisions can influence the fabric of all existence. As Einstein wrote in a letter to Lentz (1949): "My scientific work is motivated by an irresistible desire to understand the mysteries of nature and by no other feelings." (quoted from Calaprice, 2010, p. 19, own translation)

Driven by the same thirst for knowledge, the renowned philosopher René Descartes proposed the duality of mental and material reality in 1641. Based on this, we still distinguish today between subjective and objective reality. Yet these two realities do not exist in isolation from one another; instead, they interact in ways that are sometimes inexplicable and unknown.

In this regard, various approaches can be distinguished, each postulating different conceptions of how mind and matter interact. The most widely accepted is the theory of interactionism. Proponents of this approach, such as Descartes and Popper, assume that mind and matter are in constant interaction and influence one another. On the one hand, this occurs as the human mind perceives objective reality—sometimes consciously, sometimes passively; on the other hand, subjective reality influences matter through human decisions and the resulting behaviors (Gadenne, 2019, p. 382).

Human perception involves complex information processing, but also the subjective aspect of experience (Chalmers, 2017, p. 226). Humans cannot escape their subjectivity. They are inevitably confronted with the experience of what it is like to be human (Nagel, 1980, p. 322). The question is how this subjective experience can arise. How can our immaterial consciousness result from our brain, which is part of objective reality? All of our experience arises from the

physical structures of our brain, even though mind and matter share no known point of contact or clear common ground. Psychological theories of behavioral control examine how cognitive processes influence an individual's environment, with brain functions serving as the interface between mind and matter (Kuhl, 1985; Heckhausen & Gollwitzer, 1987). We are not yet able to explain exactly how this works. This ambiguity was described by Chalmers in 1995 as the "hard problem of consciousness." How the mental can influence matter remains equally unclear. Shariff et al. defined this complementary problem in 2008 as the "hard problem of free will."

To initially circumvent the fundamental questions regarding the exact possibilities of interaction between mind and matter, more abstract concepts must be employed. The Model of Pragmatic Information (MPI) operates at a higher level of description and thus does not require clarification of the detailed physical conditions (von Lucadou, 2015, p. 58). In his model, von Lucadou views the exchange between objectivity and subjectivity as an interaction process involving organizationally closed systems, in which the properties of the two forms of reality are regarded as complementary. Subjectivity is characterized by 100% Novelty (N)—that is, by autonomous information that is new to the recipient. Objectivity, on the other hand, exhibits 100% confirmation (C), which refers to temporally stable information that is already known to the recipient and to which they can link new information. Because of the complementarity of novelty and confirmation, the respective other element in subjective and objective reality assumes a value of zero.

The natural sciences place a high demand for objectivity on research, which is why measurements are conducted non-invasively. In this process, the object of measurement and the measuring instrument exist independently of one another; the instrument passively records an objective reality that is independent of it (d'Espagnat, 2006, p. 93ff). In subjectivity, a construction of reality takes place that is distorted by a wide variety of influences, with the measuring instrument and the object of measurement merging into one another. This is referred to as an invasive measurement (Maier et al., 2022, p. 22).

While elements of objective reality are replicable and intersubjectively verifiable, subjective experiences are of an individual nature and cannot be conveyed to others in the same way (Levine, 1983, p. 356).

In order for the mind to exert influence on matter, von Lucadou (2015) at the MPI postulates the necessity of pragmatic information. Because this is calculated from the multiplication of C and N, the optimal value of pragmatic information is achieved through the balanced interaction of the two components. However, the simultaneous occurrence of N and C is impossible due to their complementarity in both objectivity and subjectivity. Within the dualistic representation of the polarity of mind and matter according to Descartes (1641), pragmatic information therefore equals 0 (Kornwachs, 2022, p. 12). To account for the possibilities of the mind influencing matter, this worldview must therefore be expanded.

Bierman therefore proposed in 2001 that Cartesian dualism be overcome: He hypothesized the existence of a third form of reality alongside subjectivity and objectivity, known as “tribal reality” (p. 13). Maier et al. proposed designating this third form of reality as “sobjectivity” due to its points of intersection with subjective and objective reality (Maier et al., 2022, p. 23). Pragmatic information can exist within sobjectivity because $N > 0$ and thus C is not 100%. It follows that this third form of reality—and thus the transmission of pragmatic information in the sense of the mind influencing matter—cannot be objectively grasped.

Pauli and Jung offer another approach to explaining the interaction between mind and matter in their Unus Mundus theory, which emerged from a correspondence between 1932 and 1958 (Meier, 2013, p. 1ff). Unus Mundus describes a unified reality to which everything spiritual and material, everything psychological and physical, can be traced back (Roesler, 2016, p. 30). According to this theory, subjectivity and objectivity are interwoven on the shared, pre-material, and pre-conscious level of Unus Mundus. This level encompasses the quantum world and the unconscious as potentials and possible

realities, which is why it should also be understood as a “potential space” (Atmanspacher, 2012). Spiritual and physical reality arise from it, and through the Unus Mundus—as the shared plane of both realities—the mind can intentionally influence the material world. Atmanspacher referred to this form of synchronicity as induced correlation (2013, p. 231). In this process, objective reality is influenced by an impulse of will that distorts quantum randomness via the Unus Mundus. The process of induced correlation arises through communication with the unconscious, whose language is emotion. Therefore, a clear desire or motive to influence matter is indispensable (Iovine, 2022, p. 4).

Furthermore, Jung (1976) proposed the concept of a collective unconscious, which is shaped by archetypes. According to Jung, these are emotionally charged, unconscious, and innate patterns of perception and behavior. The collective unconscious thus represents universal human experiences, comparable to schemas that are innate to certain animal species, such as nest-building or hunting behavior.

Because the subjective-spiritual would lose its properties of autonomy through systematic influence on objective-material reality, induced correlations cannot occur systematically (Atmanspacher, 2012). It follows from this—as well as from the model of pragmatic information—that a deliberate influence of the mind on the materialization of matter is difficult to objectively detect.

Quantum theory should be consulted to derive a possible explanatory framework for these assumptions. This theory also includes the assumption that the exact position of a particle cannot be determined. Due to its wave function, we can never determine both its velocity and its current position at the same time (see Heisenberg’s Uncertainty Principle, 1927). Only the probability of a particle’s exact location at a given moment can be calculated using the Schrödinger equation (Greene, 2004, pp. 112–115), which is why this equation and the

described wave function were termed a “probability wave” by Max Born in 1926 (Born, 1926).

Before the calculation to determine a particle’s location is performed, the particle is in a so-called “superposition”; it occupies, so to speak, several potential states that overlap (Schrödinger, 1935). This state is also described by the Copenhagen interpretation, which was developed in 1927 by Bohr, Heisenberg, and colleagues and is considered the standard theory of quantum mechanics (Heisenberg, 1955, p. 143). Within the framework of the Copenhagen interpretation, asking about the exact position of a particle before measurement is meaningless, because the particle does not have a definite location at that moment (Ströhle, 2012, p. 44ff). Thus, an ontic coincidence determines the measurement result; a collapse of the wave function must occur, which is referred to as indeterminism (Greene, 2004, 117f). The particle assumes a definite state only through its observation. Building on these assumptions, Jahn and Dunne postulated in 1997 that humans influence the emergence of reality through intentional observation.

Accordingly, the mind’s influence on matter would not be limited to the production of intervening behaviors; humans could directly shape objective reality through their mental powers in accordance with their goals. Such influence involves processes of psychokinesis (PK), that is, the ability to influence objects solely through the power of the mind (Varvoglis & Bancel, 2015, p. 1). This branch of research is part of psi research, which deals with the investigation of paranormal phenomena and can be classified within the field of parapsychology. The extent of the human need to help shape the manifestation of reality through one’s own will was succinctly summarized by Maier and colleagues: “The notion of being able to materialize ideas and desires solely through the power of one’s own mind is reflected in religion, mythology, and philosophy.” (2018, p. 1, own translation).

The hypothesis of psychokinesis is based on quantum-based stochastic results that exhibit intentional observer effects (Jahn & Dunne, 1997). Furthermore,

the claim that the mind can influence the materialization of matter is further supported by a multitude of tests demonstrating the occurrence of micro-psychokinesis (micro-PK) (Williams, 2021). This refers to “tiny influences on inanimate, probabilistic systems that have effects detectable only by statistical means” (Varvoglis & Bancel, 2015, p. 1, own translation). Significant overall effects have been demonstrated in a wide variety of meta-analyses (Bösch et al., 2006; Radin & Nelson, 1989), although the unusual heterogeneity in the effect sizes of individual experiments is noteworthy. As early as the 1930s, Rhine, together with several colleagues at Duke University, conducted interesting laboratory experiments on psychokinesis. These were partly inspired by gambling scenarios and the attempt to mentally influence the outcome of such a game. Over a period of 52 years, Rhine et al. replicated 150 experimental PK tests using dice, in which volunteer participants were asked to try to intentionally influence the dice rolls. A total of 2,569 subjects were recruited, and 2.6 million dice were rolled for testing (Williams, 2021, p. 831). A small but significant overall effect was observed (Stouffer’s $Z = 18.2$, $p < 10^{-7^0}$) with an odds ratio of over one billion to one, while 31 control tests involving 153,288 dice rolls were consistent with chance in the overall results, with an odds ratio of only about two to one (Stouffer’s $Z = 0.18$, $p = .429$) (Radin & Ferrari, 1991, p. 68). The research group was thus able to demonstrate a significant effect in their PK experiments, which they further corroborated through control experiments.

Another highly relevant series of studies was conducted in the 1970s by Helmut Schmidt, known in parapsychology as the RNG (“Random Number Generator”) revolution (Williams, 2021, p. 833). Schmidt developed an electronic device capable of randomly generating two binary states with equal probability (Schmidt, 1970b). He initiated two independent test series in which a subject was seated in front of a display showing a circular array of small lights. The lights lit up individually, either sequentially in a clockwise direction or in the opposite direction, which in turn was determined by the

The output of the RNG device was determined (Schmidt, 1970a). The participants were instructed to try to consciously influence the direction in which the lights began to illuminate one after another. The underlying theoretical assumption was that the participants could mentally manipulate the RNG's output in accordance with their goals. Interestingly, the overall results suggested that the participants' attempts to influence the outcome led to contradictory results: An attempt to influence the direction in which the lights lit up yielded a success rate of 49.5% (13,695 successes out of 27,648 trials; $z = –1.55$, $p = .06$). The second test yielded similar results, which showed an even stronger tendency toward negative influence: The success rate of influencing the outcome dropped here to 49.1% (16,082 successes out of 32,768 attempts; $z = –3.34$, $p = .0004$), although during this test Schmidt specifically asked some participants to allow themselves to experience pessimistic and discouraging feelings (Williams, 2021, p. 834ff). This sequence clearly illustrates how Helmut Schmidt recognized the high relevance of the valence of directed intentions for micro-PK and utilized it to his advantage.

However, the particular significance of the experiments conducted by Schmidt lies not in their results, but in the principles employed. Even today, RNG devices play a central role in the study of micro-PK effects, particularly so-called qRNGs ("Quantum Random Number Generators"). Unlike Schmidt's RNG, QRNGs use quantum mechanical processes to generate random numbers. To ensure true randomness, phenomena such as quantum mechanical decay—which are inherently indeterministic and unpredictable—are utilized.

The influence of human intentions on quantum randomness was investigated, among other things, in a series of studies by Dechamps et al. in 2021. In this study, a qRNG was used to determine whether a positive or negative image was shown to the participant (Maier & Dechamps, 2022). Through the design of the task and subliminal reinforcement, the researchers attempted to elicit in the participants the intention to want to see positive images. In the first study, strong Bayesian evidence was found that, given the participants' intentions, the qRNG

the participants, the qRNG actually presented positive images more frequently. However, this effect diminished in the two subsequent studies, confirming instead that the results were due to quantum randomness (Maier & Dechamps, 2022).

A similar decline in the effect had already been observed in numerous studies on micro-psychokinesis (Maier & Dechamps, 2018). However, based on von Lucadou's MPI, this pattern is to be expected: an effect can only manifest during the first measurement because C < 100% at that point. As soon as C = 100%, no pragmatic information can be transmitted, which is why replication of the result is not possible.

To circumvent this problem, Maier et al. modified the design of the study by Dechamps et al. in 2022 by varying the objectivity. They divided the experiment into two conditions: In the C(max) condition, the number of positive images shown was recorded; in the C(reduced) condition, this number was discarded, and instead, the number of positive images subjectively recalled by the participants was used. In the reduced-objectivity condition, a strong micro-PK effect was observed (BF10 = 17.14). In the C(max) condition, on the other hand, a zero effect was observed. This confirms, in principle, the assumption that lower measurement objectivity favors the occurrence of micro-PK effects. However, the results could also have been due to confounding by a order effect; this was ruled out, however, by an interesting coincidence: In the C(reduced) condition, although the raw data were deleted as planned, data were later found that allowed the valence of the displayed images to be reconstructed retrospectively. On the one hand, this ruled out order effects as an explanation for the result; on the other hand, it also provided a new opportunity to examine the available data for micro-PK effects. Maier et al. restored half of the objective data found from the C(reduced) condition, while permanently destroying the other half. For the completely deleted data, a BF10 = 26.94 was observed, indicating a very strong effect in favor of H1, as well as a BF01 = 3.18, which pointed toward H0. These observations

show that maximizing objectivity, as is customary in modern natural sciences, reduces micro-PK effects (Maier et al., 2022, p. 23ff).

It can therefore be assumed that under conditions of perfect objectivity—and thus 100% confirmation—no deliberate, intentional impulse can influence matter. From this, the greatest challenge in psi research can be deduced: insufficient replicability (Jahn et al., 2000). A common occurrence in the study of micro-PK is a decline in the originally observed effect during replication. This phenomenon is referred to, among other terms, as the statistical equilibrium effect, the compensation effect, the difference effect, or the first-user effect (Dechamps & Maier, 2019, p. 411). In their 2019 publication, Dechamps and Maier also used the term “Nemesis effect,” an allusion to the Greek goddess of retributive justice. There are various explanatory approaches for this effect, including patterns of selective publication, an increasing quality of the experiment, and several psychological variables, among them practice effects.

According to Dechamps and Maier (2019, p. 414), the frequently observed interplay of effect and counter-effect in psi research leads to a systematically recurring pattern in the temporal development of the effect. A 2018 publication by the two researchers, which examined the results of two micro-PK studies on the intentional influence of smokers and nonsmokers on the frequency of presentation of neutral images and addiction-related images, plays a particularly important role here. With regard to these studies, Dechamps and Maier proposed a method to examine the characteristic pattern of variation in the effect more closely. They assumed that the observed effects, in a systematic interplay of effect and counter-effect, would exhibit the pattern of a harmonic oscillation. The function derived from this was intended to allow for a prediction of the values for the next 200 participants. Although the data collected subsequently were close to the predicted trend, no statistical significance was found (Dechamps & Maier, 2019, p. 423), although

these findings at least suggest the potential relevance of developing such a predictive curve.

Despite these particular challenges in psi research, Cardeña was able to find clear and sufficient evidence for the existence of such phenomena following an extensive meta-analysis: “The evidence provides cumulative support for the reality of psi that cannot simply be dismissed on the grounds of study quality, fraud, selective reporting, experimental or analytical incompetence, or other common criticisms. The evidence for psi is comparable to that for established phenomena in psychology and other disciplines, although there is no consensus on its nature.” (Cardeña, 2018, p. 1).

The study that follows is also intended to contribute to expanding our knowledge of psi phenomena.

# Methods

## Research Project

As part of an empirical practicum supervised by Prof. Dr. Markus Maier, I worked with Julia Berner and Christoph Bäck to design an experiment intended to elicit and record the occurrence of micro-PK. The study investigated whether a specific outcome of a random event occurs significantly more frequently when the participant wishes for it to happen. The basic idea behind our experiment is thus reminiscent of Rhine's early experiments, which were based on gambling scenarios. We specifically sought to intrinsically motivate the participants to intentionally influence a random experiment.

Building on the promising results of this study, I repeated the experiment to recruit additional participants. The study design remained unchanged, allowing the data from the newly recruited participants to be combined with the previous dataset and analyzed using Bayesian methods. For simplicity, I will refer to the original experiment below as the 1st study run / 1st data collection and the expansion as the 2nd study run / 2nd data collection.

## Operationalization

The randomized experiment—which could potentially be influenced by intentional observation—was conducted in the form of a card draw. Participants were instructed to draw one card from two face-down cards (red and black). The aim was to investigate whether—and to what extent—a specific incentive to reveal the red card could influence its probability of occurrence. The number of red cards revealed served as the dependent variable, while the independent variable consisted of the incentive to reveal the red card.

This specific incentive was intended to intrinsically motivate the participants and specifically elicit the intention to reveal the "correct" (i.e., red) card. To this end, the participants were presented with a story in two parts: The first part was shown at the beginning

, while the second part could only be viewed by drawing the red card in the randomized experiment.

In addition to the experimental group, there was also a control group in which no specific incentive to reveal the red or black card was provided.

For exploratory purposes, several self-assessment questionnaires were additionally used in the original experiment. These were used to measure autonomy, available self-control capacity, and vitality; items from the Implicit Theory of Willpower for Strenuous Mental Activities Scale were also included. In the second survey, only a questionnaire measuring vitality was used.

**Hypotheses**

H1: In the experimental group, the number of red cards revealed is significantly higher than in the control group.

H2: The number of red cards revealed in the experimental group is higher than would be expected by chance.

**Data Collection**

It was an online experiment. The first round of the study was conducted in collaboration with Julia Berner and Christoph Bäck conducted, where we jointly recruited the participants. Recruitment took place via social media and through the distribution of flyers over a period of 7 weeks. For the second round of the study, which was conducted without the involvement of my former research group, I used LMU's SONA system to recruit participants in addition to the methods mentioned above. This is a cloud-based online system for organizing studies, which offers student participants an incentive to participate.

Because the data were analyzed using Bayesian statistics, the participants from the second survey could be easily added to the first dataset, thereby enabling a joint analysis of the collected data.

**Sample**

For the first study run, we were able to recruit 298 participants, 13 of whom had to be excluded because they stated that they had not taken the study seriously. Two additional participants had to be excluded because they were still minors. Consequently, 283 participants remained for analysis.

For the second round of the study, I was able to recruit an additional 343 participants, 13 of whom had to be excluded because they had not taken the study seriously or because they were minors. By merging the two datasets, a total of 613 participants were included in the analysis. Table 1 presents a descriptive statistical analysis of the age distribution of all participants in the combined dataset. Table 2 presents a descriptive analysis of their gender distribution.

**Table 1**

*Combined Dataset, Age Distribution in the Sample*

| | Mean | Standard Deviation | Minimum | Maximum |
|---|---|---|---|---|
| Age | 27,938 | 12,955 | 18 | 87 |

**Table 2**

*Combined Dataset, Gender Distribution in the Sample*

| | Male | Female | Diverse | Total |
|---|---|---|---|---|
| Frequency | 165 | 442 | 6 | 613 |

**Procedure**

The study procedure varied depending on whether the participant was randomly assigned to the experimental or control group. However, both groups received information regarding data protection, voluntary participation, and the right to withdraw from the study, and were asked to provide their gender and age.

In the control group, participants were directed straight to the randomization experiment without first receiving an incentive to reveal the red card. In the experimental group, however, to optimize the design of the incentive, participants could first choose the topic they found more interesting from the two categories “Reincarnation/Rebirth” and “True Crime/Criminal Case.” The choice of topic area determined which story was presented to the participant as an incentive.

If the “Reincarnation/Rebirth” topic area was chosen:

*“The Case of James Leininger: James Leininger claimed to have memories of a past life. As early as age 2, he repeatedly had vivid dreams about a plane crash and claimed to be an American pilot who had been killed by the Japanese. It turned out that James did indeed possess detailed knowledge about the life of an American pilot from World War II. This presented a major mystery to his family and the scientific community. But what specific details did James know that drew so much attention to the boy? To find out, try to reveal the red card in the following game. On the back, you’ll find the key to the James Leininger case. If you succeed, you’ll learn his entire story.”*

When selecting the “True Crime/Criminal Case” category:

*“When I died, I wasn’t yet a fully formed person, but I was already someone. I was killed for who I was. Did I die alone? No, there were six of us, in the same house at the same time. Now no one lives there anymore—not my mother, not my two fathers.*

*The crime scene: A farm in the open countryside. The night was lit by the moon, yet no one saw anything. What was found? Footprints leading to the farm, but none leading away; all the victims were hidden. And the animals? Even four days later, they were still calm, fed, and cared for. Well, can you find my killer? To find out, try to turn over the red card in the following game. On the back, you'll find the missing details you need to unmask the killer."*

These text passages were the incomplete first part of each story. To receive the second part and learn the ending of the story, the participants were asked to try to turn over the red card in the subsequent randomized experiment. Once the red card was turned over, the participant was presented with the ending of their story for the selected topic area.

If the topic "Reincarnation/Rebirth" was selected:

*"Congratulations. You have revealed the red card and thus gained access to the missing information in the case of James Leininger.*

*James Leininger claimed to have memories of a past life as a pilot in World War II and a plane crash. He was able to provide precise details, such as specific information about the aircraft, the carrier group, the name of his comrade, and even personal details about his family at that time. James's parents discovered that his statements bore striking similarities to the death of a World War II pilot named James Huston. Furthermore, the knowledge James seemed to possess about this deceased pilot exceeded what would normally be expected of a young child. There is no definitive proof that James Leininger actually has memories of a past life. It is entirely possible that this is a hoax. However, the idea of reincarnation should not be ruled out when investigating such cases."*

When selecting the “True Crime/Criminal Case” topic:

*“Congratulations. You’ve revealed the red card and thus gained access to the missing information needed to identify the murderer.*

*Well, did you find my killer? I am Josef Gruber, born in 1919, died in 1922—still a baby, yet already with a story. I died one night in March on my family’s farm. They were all bludgeoned to death that night—my two grandparents, my mother, my sister, even the maid. Six innocent lives, you might think… But none of us were quite so innocent. I was the result of an affair: between my grandfather and my mother. My official father, however, was a young man from the village. He loved me very much until he found out he wasn’t my biological father. Then he hated us all. One night in March 1922, he stormed into the farmyard, killed us all, lived under the same roof with our bodies for four days, and tended to the livestock. No one realized we were already dead. But he didn’t want to see us anymore, so he hid us. Then he left the farm. That was my story. One more note: To this day, no one knows who the perpetrator really was, because the case remains unsolved even 101 years later, and my story here is pure speculation. Thanks for joining me in solving the mystery!”*

If the red card was not revealed, the participant could not be shown the end of the story. The experiment ended at this point. Upon completion of the experimental portion of the online study, both the control group and the experimental group were given a series of questionnaires.

In the study, participants were asked to self-report on various characteristics. The questionnaires included items measuring autonomy, available self-control capacity, vitality, and items from the Implicit Theory of Willpower for Strenuous Mental Activities Scale. In the second round of the study, however, only vitality was assessed.

At the end of both rounds, participants were asked, in a “hand-on-heart” question, whether they had taken the study seriously. Regardless of the participant’s answer, they were thanked for their participation, and a brief explanation of the study’s background was provided.

**Data Analysis**

The JASP program was used for data analysis. Bayesian statistics were also employed. This form of statistical analysis is flexible and cumulative, allowing new participants to be integrated into a dataset that has already been analyzed.

For both runs individually, as well as for the combined dataset, I first examined the data purely descriptively. Subsequently, a chi-square test was conducted within the Bayesian contingency table, for which a prior of 1 was set. Additionally, I applied a Bayesian t-test for independent samples, with a defined Cauchy prior width of 0.1.

In the post hoc analysis, I also conducted a sequential analysis based on the Bayesian t-test for independent samples to examine the effect pattern across the number of participants included. Furthermore, I applied a Bayesian one-sample t-test to investigate the effects of gender.

## Results

Note: The variable “correct” represents the color of the revealed card. “1” means that a red card was revealed; “0” means a black card.

### 1. : Original Experiment

The descriptive data analysis indeed confirmed the expected effect: When given an incentive to turn over the red card (in the experimental group), it was turned over more often than when no incentive was given (in the control group) and more often than would be expected by chance. In the control group, descriptive statistics showed—as expected—a distribution of revealed red and black cards that would also be expected by chance (see Table 3).

**Table 3**

*1. Survey, descriptive statistical analysis of the relationship*

| Condition | correct | Absolute Frequency | Relative Frequency |
|---|---|---|---|
| Incentive | 0 | 64 | 45,714 |
| | 1 | 76 | 54,286 |
| | Total | 140 | 100,000 |
| No incentive | 0 | 71 | 49,650 |
| | 1 | 72 | 50,350 |
| | Total | 143 | 100,000 |

Building on these promising observations, we used the Bayesian t-test for independent samples to determine whether the descriptively observed result was also associated with a statistically significant difference in the means between the experimental and control groups. To assess this, we tested the alternative hypothesis H1. Specifically, we determined whether, given a specific incentive, significantly more red cards were revealed than in the control group without such an incentive. This analysis yielded a Bayesian factor (BF) of

0.848, meaning there was no evidence for our alternative hypothesis (H1). A BF > 1 would have supported the acceptance of our alternative hypothesis.

To test H2, we then used the Bayesian contingency test. This was intended to show whether the observed frequencies of red cards deviated from the frequencies expected by chance—that is, whether red cards were drawn more frequently in the experimental group than would be expected by chance. However, no significant deviation was found (BF = 0.091).

**2. Data Collection**

Interestingly, a descriptive analysis of the data from the second study run revealed an effect opposite to that found in the descriptive statistical analysis of the
first study run: The number of red cards revealed decreased when the incentive was present; in the experimental group, more black cards than red cards were revealed. In the control group, as would be expected by chance, there was an even distribution of red and black cards (see Table 4).

**Table 4**

*2. Survey, descriptive statistical analysis of the relationship*

| Condition | correct | Absolute Frequency | Relative Frequency |
|---|---|---|---|
| Incentive | 0 | 81 | 55,102 |
| | 1 | 66 | 44,898 |
| | Total | 147 | 100,000 |
| No incentive | 0 | 90 | 49,180 |
| | 1 | 93 | 50,820 |
| | Total | 183 | 100,000 |

The Bayesian t-test for independent samples used to test H1 yielded a BF of 0.340 for the second survey. Here, too, the null hypothesis should therefore be accepted. In the Bayesian contingency test, which was designed to examine the assumptions of H2

, no confirmation of the alternative hypothesis was found (BF = 0.414).

**Combined “ ” dataset**

The combination of data obtained from the first and second study rounds is also relevant for data analysis. A joint descriptive analysis revealed an even distribution of the red and black cards revealed in both the experimental and control groups, as would be expected by chance (see Table 5).

**Table 5**

*Combined Dataset, Descriptive Statistical Analysis of the Relationship*

| Condition | correct | Absolute Frequency | Relative Frequency |
|---|---|---|---|
| Incentive | 0 | 145 | 50,523 |
| | 1 | 142 | 49,477 |
| | Total | 287 | 100,000 |
| No incentive | 0 | 161 | 49,387 |
| | 1 | 165 | 50,613 |
| | Total | 326 | 100,000 |

Thus, the incentive in the combined dataset does not appear to affect the probability of a red card being issued.

The Bayesian t-test for independent samples yielded a BF of 0.396, so H1 could not be accepted. H2 could not be confirmed for the combined dataset either: The Bayesian contingency test yielded a BF of 0.129.

The assessment of vitality was purely exploratory and revealed no noteworthy findings regarding the relationship under consideration.

## Discussion

In micro-parapsychology research, attempts are often made to induce an intentional influence by participants on quantum randomness through extrinsic motivation (Maier, 2018). In contrast, this study design was based on the attempt to intrinsically motivate participants to influence matter.

Because Bayesian statistics allow for the combination of data sets from different measurement periods, it was possible to conduct an insightful analysis of individual measurements, compare them, and consider them collectively. No significant effect emerged from any single measurement point, although certain anomalies were certainly noticeable from a purely descriptive perspective.

“The problem with parapsychologists, however, is that they do not accept ‘no’ for an answer.” (Eisenbud, 1992, p. 161, cited in Braude, 2016, own translation)

Despite the lack of statistical significance in my measurements, I would like to examine the observed trends in the datasets more closely. It became apparent that they differ significantly from one another: While the data from the first survey period at least descriptively supported H1 and H2, the results from the second survey deviated from the hypotheses even in a descriptive analysis. When considering both datasets together, our assumptions were likewise not confirmed; here, chance alone seemed to determine the occurrence of the red and black cards. It appeared as though the paranormally observed effect from the first survey had simply leveled off in the replication study, so that no deviation from a random distribution was noticeable when the datasets were combined. Thus, I was unable to replicate the observed effect from the first study by including additional participants, let alone demonstrate an amplified effect and achieve statistical significance.

However, such developments are not isolated cases in micro-PK research. The decline in originally observed effects when attempting replication is a common problem in psi

research (Bem et al., 2001, p. 208). As von Lucadou noted in 2007: “Whenever a psi experiment initially shows positive results, subsequent data or replications will erase the primary observed effect…” (p. 52, own translation).

The most famous example of this decline effect is the large-scale micro-PK study by the PEAR group (Jahn et al., 1987). In that study, the measured effect size decreased continuously with each replication attempt (von Lucadou, 2015, p. 5). Similarly, the data from two studies by Maier and Dechamps (2018) initially showed a strong micro-PK effect, which declined significantly in the subsequent replication study (p. 289). In these studies, Maier and Dechamps divided their participants into two subgroups: smokers and nonsmokers. They expected smokers to influence the output of a qRNG that could generate either a smoking-related or a neutral image. For the non-smokers, the researchers assumed a random distribution of neutral and smoking-related images. In the first round of the study, these assumptions were confirmed by a strong effect; however, this was no longer the case in the second round. An initially clear effect of smokers’ influence on the qRNG’s output disappeared completely in the replication study, although both studies provided moderate to strong evidence that there was no deviation from a random distribution in the occurrence of images within the group of non-smokers (Maier and Dechamps, 2018, p. 290). In general, the effects observed in these studies resemble the trends I observed in my replication study. While the control group maintained its random distribution in the replication study as well, the effect observed descriptively in the experimental group changed its orientation completely.

Of course, one can conclude from these results that no actual, stable micro-PK effect occurred. Scientific standards also suggest this interpretation. However, the use of Bayesian statistics may also have been a reason why no clear effect with statistical significance could be demonstrated. In principle, Bayesian statistics are considered

extremely conservative. The chosen prior of 1, which does not favor any specific prior assumptions, and the Cauchy prior width of 0.1, which allows for moderate precision and robustness against outliers, represent quite conservative choices within Bayesian statistics.

However, the observed trajectory of the effect also provides an indication of recurring, systematic patterns of variation in micro-PK effects. As mentioned earlier, Maier and Dechamps assumed that this compensatory effect is subject to a systematic interplay of effect and counter-effect, the modeling of which could allow for the prediction of future results (Maier & Dechamps, 2019, p. 414). Although they were unable to achieve statistical significance for their trajectory model at the time (p. 423), their assumptions appear to be quite relevant to my observations. Given the large number of experiments in the field of psi research that appear to be plagued by a similar decline in originally observed effects (Varvoglis & Bancel, 2015, p. 272), a closer examination of the possible causes for this decline is warranted.

Theoretical models can be used for this purpose: The Model of Pragmatic Information by von Lucadou (2015) offers an approach for post-hoc analysis of the development of the effects in the present data. The model assumes that micro-PK effects disappear when additional data are collected, as was the case in this study. This assumption also follows from the author's statement: "…a strict replication is the best way to destroy the effect." (von Lucadou et al., 2007, p. 61). First, to explain this phenomenon, we can consult the formula for calculating the pragmatic information required, according to the MPI, for a mental influence on matter. As discussed earlier, a replication study may involve no novelty or only reduced novelty, which makes a balanced occurrence of N and C impossible and thus inevitably reduces or eliminates the observable influence of mind on matter in the replication study.

Furthermore, the diminishing effect can also be explained on the basis of the two laws of the MPI: The first law states that psi phenomena are non-local entanglement correlations in complex, self-organizing systems, which are induced by pragmatic information. The second law states that the correlation disappears as soon as such a non-local entanglement correlation is used as a signal transmission (von Lucadou, 2015, p. 7). These two laws can be applied to the circumstances of replication studies: As soon as an attempt is made to systematically isolate psi effects and replicate them, their non-causal and spontaneous nature is disrupted. Replication studies, in which the conditions of the original experiment are precisely replicated, presuppose that psi effects are understood as causal signals that can be reproduced. According to the second law of the MPI, this causes the effect to disappear (von Lucadou, 2015, p. 20).

Furthermore, it can be inferred from the theoretical derivations of a research group consisting of von Lucadou, Römer, and Walach (2007) that a violation of the probability rule in quantum mechanics would conflict with the “no-signal” theorem (NT axiom in generalized quantum theory). This theorem is intended to prevent physical paradoxes and states, for example, that no signal can travel faster than light. To avoid this conflict, the systematic detection of micro-PK effects—and the use of these effects as a signal—must be prevented. This makes the successful replication of observed micro-PK effects extremely unlikely (Jakob et al. 2020, p. 229).

The second law of thermodynamics must also be invoked to explain the decline effect. The force that appears to act antagonistically in many replication studies in the field of psi research—preventing the unequivocal replicative detection of micro-PK effects through systematically occurring counter-effects—could be related to the second law of thermodynamics. This law states that entropy in closed systems must always increase. Thus, it is established that information does not arise from randomness and that natural processes lead to an increase in disorder

(Adamenko, 1989, p. 110). Under the influence of micro-PK, a chaotic system—such as a random quantum event—would become ordered without the input of energy. Consequently, growing evidence for micro-PK effects is assumed to be the cause of an entropic counter-effect. Dechamps and Maier (2019, p. 414) posited that this counter-effect occurs as soon as a certain threshold of experimental evidence for mentally induced deviations from quantum randomness is reached. Once the originally observed effect is successfully attenuated by the entropic counter-effect, the counter-effect itself also subsides. Thus, the micro-PK effect can reappear. Accordingly, the balancing interplay between the counter-effect and the effect should manifest itself in a specific pattern comparable to a damped harmonic oscillation (Jakob et al., 2020, p. 230), and it can be assumed that this course of the effect is predictable and detectable (Maier & Dechamps, 2018, p. 292).

However, there are also alternative explanations for the significant decline in the effect during the second survey: When recruiting additional participants, I took care not to include anyone who had already been surveyed in the first survey. However, it cannot be ruled out that some participants from the first survey also took part in the replication study. This could account for the occurrence of fatigue and training effects. In the dice studies conducted by Rhine and his colleagues in the early 1930s, strong effects were observed in the initial trials, which diminished as the study progressed. The research group attributed this trend to a decline in the subjects' motivation as the study duration increased. While the subjects were still motivated and excited during the first runs, they became increasingly discouraged by their failures (Varvoglis & Bancel, 2015, p. 12). In the studies by Rhine et al., the participants were discouraged by unsatisfactory dice rolls; in the case of my investigations, this may have been due to revealing the wrong card in the first trial.

It is important to emphasize here that when intentions influence the manifestation of reality, the valence of the intention also plays a decisive role. In general, it appears that a relaxed yet focused state of mind allows for a more productive influence on matter than a strained state characterized by a tense focus on performance and success (Varvoglis & Bancel, 2015, p. 12). In his earliest experiments, Schmidt deliberately induced stress in the subjects and was able to demonstrate that, under stress, the influence on matter was opposite to the actual intention (Schmidt and Pantas, 1972). Schmidt observed the same effect in his experiments with the RNG among participants with a pessimistic attitude (Williams, 2021, p. 834ff). The sample from the second survey in my study may have felt particular pressure to successfully reveal the red card. This influence of negative emotions could potentially have led to a reversal of the effect.

Furthermore, given the assumptions made, it is conceivable that the study results were influenced by the researchers' intentions and desires. While my research group's confident and unbiased intentions in the first survey made the occurrence of micro-PK possible, my strong desire for a positive result in the second survey—and the accompanying stress—may have negatively influenced the potential space.

However, other psychological processes must also be considered: While publication bias and a change in the quality of the study did not play a role in this case, the subjects' unconscious attitudes toward paranormal phenomena may well have had an influence. In principle, the participants were not given any information prior to their participation regarding the exact background of the study that would have allowed them to infer the subject area of parapsychological research. However, because the second survey also included students recruited via the SONA system who are familiar with Prof. Dr. Maier's research, their personal opinions about this field of research may have influenced their attitude during the

conduct of the study. Implicit, negative beliefs regarding the study's research field could have played a decisive role in the direction or nature of their intentional influence on the random event (Dechamps & Maier, 2019, p. 411).

Whether due to the participants' implicit rejection of the field of research or due to the researchers' intentions: it is possible that humans themselves, through their own uncertainties and fears, stand in the way of proving supernatural effects. In 1992, Eisenbud proposed the hypothesis that the unconscious human motivation to live in an orderly world may be responsible for the decline in paranormal effects (Braude, 2016).

**Limitations**

This study was a student project. It began as part of an empirical internship in the form of group work; subsequently, additional participants were recruited individually for my bachelor's thesis. Such projects are limited in scope, although the inclusion of additional participants in the second round of data collection allowed for a reasonably adequate sample size (n=613) to be achieved overall.

The experiment was conducted online, although online studies are generally not ideal for determining micro-PK effects. To enable the mind to exert a targeted influence on the manifestation of reality, the attentive cooperation of the participants is essential. Only in this way can the participants' intentions develop with the precision required by the given task. In online studies, however, there is little opportunity to control for confounding factors arising from distracting environmental conditions or a lack of concentration during the task.

It might also have been advantageous to formulate and test non-directional hypotheses. Micro-PK research certainly includes a large number of promising meta-analyses that suggest the occurrence of substantial effects (Bösch

et al., 2006). However, these effects often manifest unsystematically; they disappear and reappear in different forms and seem to be elusive in nature (Dechamps & Maier, 2019, p. 209). Testing non-directional hypotheses can therefore be advantageous: the counter-effect from the second survey would thus have been placed in a different context.

The two-part stories designed to create an incentive in the experimental group also warrant scrutiny. It is unclear to what extent the opportunity to complete the story could have intrinsically motivated the participants. After all, a representative population survey conducted in the summer of 2018 (Stiftung Lesen & Institut für Demoskopie Allensbach) showed that 64% of the 1,295 respondents generally found reading easy, and 84% of these individuals also stated that they enjoyed reading (p. 15f). This suggests that the incentive to complete a story could indeed intrinsically motivate a large portion of the population. After all, the participants were also able to choose the topic they found more interesting from two different subject areas. However, educational level also has a major influence on reading behavior: The survey showed that people with a higher level of education are more likely to enjoy reading and also find it easier (p. 18). Thus, the choice of a reading task designed to foster intrinsic motivation may have inadvertently incorporated the influence of the participants' educational levels into the data.

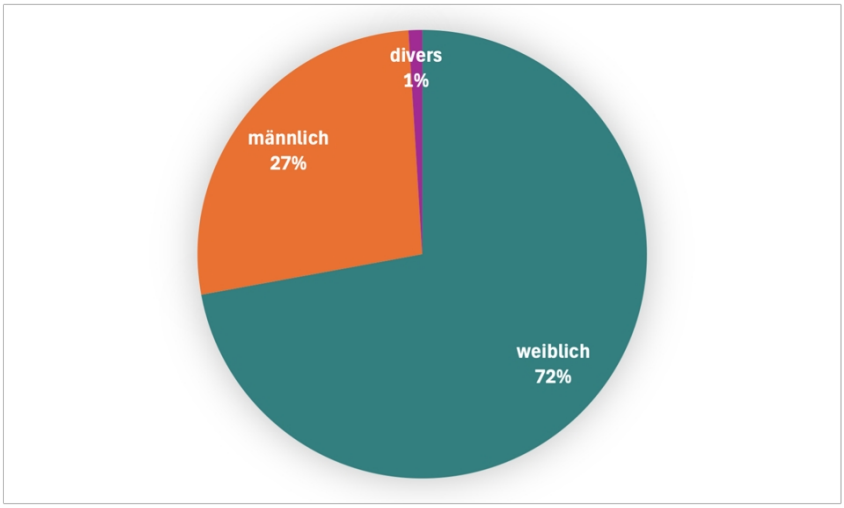
Another point of criticism is the uneven allocation of participants to the experimental and control groups. The division of participants was done automatically on a random basis. In the combined dataset, however, a total of 326 participants were assigned to the control group, while only 287 were assigned to the experimental group. If the participants had been distributed evenly, the experimental group could have been significantly larger, which might have resulted in a stronger effect in the statistical analysis.

Furthermore, it could be criticized that the participants in the control group were also instructed to reveal the red card. Although they were not given a specific incentive to do so, their task description included the following instruction: “Please try to find the red card.” For some participants, this instruction may have been incentive enough to reveal the red card. In this way, intentional influence on the card selection could also have been induced in the control group.

Furthermore, the overrepresentation of women in the sample must also be addressed. In all three datasets (1st study run, 2nd study run, combined dataset), the sample is characterized by an uneven gender distribution: the majority of participants were women. This is particularly evident in the combined dataset (see Figure 1).

**Figure 1**

*Combined dataset, gender distribution in the sample*




There may be various reasons why more women participated in the study. For example, the predominantly female social environment I used to recruit participants. Another possible explanation is the majority of female students in the psychology department, who were primarily recruited to participate via the SONA system. Furthermore, a lack of interest among men in psychology as a scientific discipline may also account for their sparse participation

Explain: A study from the United Kingdom showed that, on average, men perceive psychology as less relevant (Mercer, 2013). This could mean that men are less likely to participate in psychological studies. The lack of representativeness in the sample is certainly relevant, as the ability to perform micro-PK is potentially gender-dependent. In a study by Mossbridge and Radin (2021), strong gender differences were observed in performance on psi tasks (p. 106). The research team demonstrated that men and women employ different strategies when performing micro-PK (p. 99). Furthermore, the results of this study suggest that belief in psi effects is positively and independently predicted by gender ($p < 0.000003$). People who identify as female are more likely to trust their psi abilities (Mossbridge & Radin, 2021, p. 102).

Based on these results, Mossbridge and Radin concluded with regard to psi research: "…the effects of gender should not be ignored." (2021, p. 106). Therefore, I will now examine in more detail the effects and influences of gender on the results of my study.

Note: Due to the small number of participants who could not identify as either female or male, I have excluded them from my analysis.

**e Influence of Gender**

In this study as well, the gender of the participants appeared to have influenced their intentional attempts to influence the random experiment. First, I would like to discuss the gender differences that emerged during the first data collection. If we consider only the data from the women in this survey from a descriptive perspective, the distribution in the experimental group is as would be expected by chance. In the control group, however, the black card was revealed significantly more often (see Table 6).

**Table 6**

*1. Survey, descriptive statistical analysis of the relationship for female participants*

| Condition | correct | Absolute Frequency | Relative frequency |
|---|---|---|---|
| Incentive | 0 | 52 | 49,524 |
| | 1 | 53 | 50,476 |
| | Total | 105 | 100,000 |
| No incentive | 0 | 51 | 53,684 |
| | 1 | 44 | 46,316 |
| | Total | 95 | 100,000 |

This observation differs from the results based on the complete dataset from the first survey: Descriptive statistics actually showed the expected effect of a higher-than-chance frequency of red cards under the given incentive, while the occurrence of both cards in the control condition was distributed as would be expected by chance.

In the first survey, the men showed a positive effect in both the control and experimental groups, meaning they revealed the red card more frequently (see Table 7).

**Table 7**

*1. Survey, descriptive statistical analysis of the relationship for male participants*

| Condition | correct | Absolute Frequency | Relative frequency |
|---|---|---|---|
| Incentive | 0 | 11 | 33,333 |
| | 1 | 22 | 66,667 |
| | Total | 33 | 100,000 |
| No incentive | 0 | 19 | 40,426 |
| | 1 | 28 | 59,574 |
| | Total | 47 | 100,000 |

Although a descriptive analysis of all participants in the control group reveals a balanced distribution of red and black cards, gender-specific analysis reveals deviations that are greater than expected by chance. A systematic opposite trend appears to cause these deviations to balance out in the overall data of the control group, resulting in an apparently random distribution.

Interestingly, in the second survey, women in the experimental group showed, from a descriptive statistical perspective, an apparently above-chance frequency of the black card, just as in the control group (see Table 8).

**Table 8**

*2. Survey, descriptive statistical analysis of the relationship for female participants*

| Condition | correct | Absolute Frequency | Relative frequency |
|---|---|---|---|
| Incentive | 0 | 63 | 55,263 |
| | 1 | 51 | 44,737 |
| | Total | 114 | 100,000 |
| No incentive | 0 | 69 | 53,906 |
| | 1 | 59 | 46,094 |
| | Total | 128 | 100,000 |

In the control group of the second survey, men turned over more red cards than black cards (see Table 9). Again, it should be noted that when both genders were considered together in the control group, the distribution of the cards was as would be expected by chance. Here, too, there appears to be a balancing mechanism that shifts the unsystematic gender-specific deviations toward the presumed random distribution.

The descriptively random distribution of the occurrence probabilities for the two cards in the experimental group—as was the case for women in the first dataset—is now observed among men (see Table 9).

**Table 9**

*2. Survey, descriptive statistical analysis of the relationship for male participants*

| Condition | correct | Absolute Frequency | Relative frequency |
|---|---|---|---|
| Incentive | 0 | 16 | 51,613 |
| | 1 | 15 | 48,387 |
| | Total | 31 | 100,000 |
| No incentive | 0 | 21 | 38,889 |
| | 1 | 33 | 61,111 |
| | Total | 128 | 100,000 |

Thus, even within the same gender, there were dramatic differences between the two surveys. Mossbridge and Radin also made this observation in a gender difference analysis of their study on psi performance as a function of demographic and personality factors in smartphone-based tests (2021): The second dataset revealed a gender effect that acted in the opposite direction to the effect found in the first dataset (p. 89).

Similarly, the data from my study also revealed a marked shift in the influence of gender between the two survey time points. Based on the analyses conducted so far, it is unclear how these shifting gender effects will manifest in the joint analysis of the two datasets. To investigate this, we must now examine potential gender effects in the combined dataset from the first and second surveys.

It appears that, when considering the combined dataset from a purely descriptive perspective, women in both the control and experimental groups turned over the black card more frequently, as was also the case in the second survey (see Table 10). This above-chance frequency of the black card in both groups was not expected, at least based on the hypotheses established for this study.

**Table 10**

*Combined dataset, descriptive statistical analysis of the relationship for female participants*

| Condition | correct | Absolute frequency | Relative frequency |
|---|---|---|---|
| Incentive | 0 | 115 | 52,511 |
| | 1 | 104 | 47,489 |
| | Total | 219 | 100,000 |
| No incentive | 0 | 120 | 53,812 |
| | 1 | 103 | 46,188 |
| | Total | 223 | 100,000 |

Men, on the other hand, revealed the red card more frequently under both conditions; in the control group, there was even a comparatively greater deviation from chance (see Table 11).

**Table 11**

*Combined dataset, descriptive statistical analysis of the association for male participants*

| Condition | correct | Absolute Frequency | Relative frequency |
|---|---|---|---|
| Incentive | 0 | 27 | 42,188 |
| | 1 | 37 | 57,813 |
| | Total | 64 | 100,000 |
| No incentive | 0 | 40 | 39,604 |
| | 1 | 61 | 60,396 |
| | Total | 101 | 100,000 |

The men thus showed a positive effect, regardless of whether they were given a specific incentive to reveal the red card. To test the statistical significance of this observation, the Bayesian one-sample t-test

. This test allows us to determine whether the mean of a sample differs significantly from a specific value. In the case of my study, it thus enables us to examine in the post-hoc analysis whether, for men or women, the frequency of red cards differs significantly from a random distribution (50%) when considered as a whole.

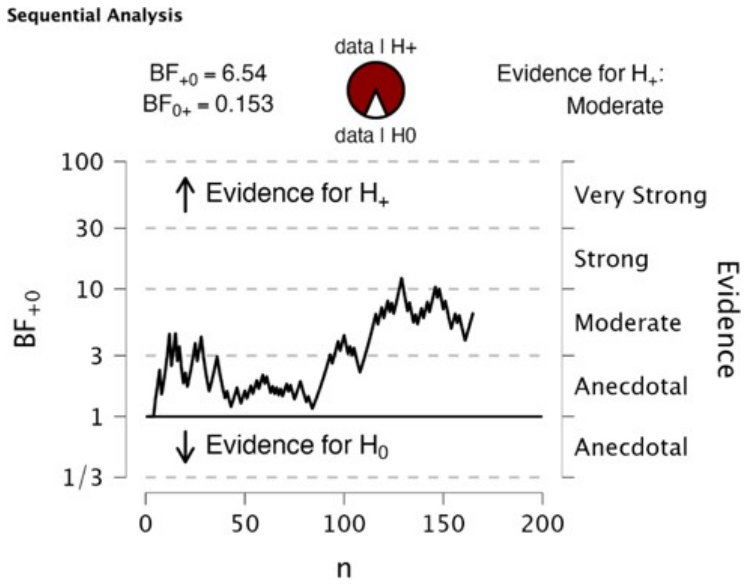
I derive the post-hoc assumption for this test from the existing hypotheses regarding the occurrence of micro-PK in this study and assume that the number of red cards exceeds 50%.

As expected, a low BF of 0.157 was observed for women here, suggesting that the result is purely due to chance. This result was also confirmed by the calculated 95% Bayesian confidence interval: [0.422; 0.515]. An exploratory test to determine whether the number of red cards for women generally lies below 50% revealed a stronger, though still low, BF of 1.129.

For men, however, a significant BF of 6.540 was calculated, indicating that the number of red cards exceeds the random distribution. This thus provides moderate support for the hypothesis that there is a significantly above-chance deviation from randomness and a positive effect independent of the incentive. This is also reflected in the 95% Bayesian confidence interval: [0.518; 0.670]. It is possible that the male participants in the control group were intrinsically motivated to intentionally influence the card draw simply by the instruction to reveal the red card. In the sequential analysis (see Figure 2), it can also be observed that the BF for this calculation had at times even reached a value just above 10. In psi research, this is considered strong evidence in favor of H1 over H0 (Wagenmakers et al., 2011). For the men, therefore, a temporarily significant effect was indeed demonstrated in the combined analysis of the two datasets, suggesting that the number of red cards significantly exceeded the random distribution.

**Figure 2**

*Combined Dataset, Sequential Analysis for Male Participants*




A comparable gender difference in psi research was also observed in an analysis of the 12-year experimental PEAR program by Jahn and colleagues (1997), which comprised over 1,000 experimental series. Among men, significantly stronger overall correlations were found between the participants' intentions and the shifts in mean values (p. 356). Approximately 66% of male operators succeeded in intentionally influencing the outcome of random binary processes in the direction they intended, whereas among women, the figure was only 34%. In addition, Dunne conducted a particularly extensive study of gender differences in the PEAR Program's database in 1998. She, too, was able to show that men's average results correlated significantly more strongly with their previously recorded intentions (p. 3).

Should we therefore assume that men are generally more capable of micro-PK? A report by Bem (2003) presents contrary results: The psi effects observed in this study could be attributed entirely to the female participants. Bem explained this finding by citing women's higher emotional reactivity (p. 9).

To date, the literature in the field of psi research provides no basis for assuming systematic gender differences in the ability to perform micro-PK (Bem,

2003, p. 9; Williams, 2021, p. 876ff). In order to make well-founded statements about the influence of gender on psi effects, further concrete investigations of this relationship are necessary.

Nevertheless, to develop a framework for explaining the gender differences observed in my study, I would like to refer to Heider's (1958) attribution theory and Weiner's (1985) elaborations in the form of the three-dimensional classification scheme. This theory suggests that people can attribute failures and successes to various factors. These factors can be controllable or uncontrollable, internal or external, and stable or variable. Meece et al. explained in 2006 that men and women exhibit different attribution patterns. While men associate their successes significantly more often with their internally stable abilities, women are more likely to cite their supposedly inadequate internally stable abilities as the cause of their failures (Meece et al., 2006, p. 354; Adenstedt, 2021, p. 78). This reflects a lower level of confidence among women in their own competencies.

As previously discussed, a person's emotions play a decisive role in intentionally influencing the way reality unfolds (Iovine, 2022, p. 4). Low confidence in one's own ability to influence a random experiment could thus stand in the way of successful intentional influence. This could explain why, in the course of my study, women appeared to be less able to intentionally influence the card draw.

**Outlook**

“Nature shows us only the lion’s tail. But there is no doubt in my mind that the lion is there, even if its size prevents it from revealing itself immediately to the eye.” (Einstein in a letter to Zangger, 1914, cited in Klein et al., 1993, p. 602)

It is evident that extensive research in the field of psi research is still needed to gradually uncover the mysteries of paranormal influence on objective reality. For example, in-depth studies on gender differences in the practice of micro-PK are warranted.

Based on the current state of research, including insights from quantum physics, it can be assumed that intentional influence on seemingly random events in the material world must be possible. Empirically capturing this could be one of the greatest challenges of our time. At present, the primary task for the field of psi research is to explore possible approaches to studying micro-PK, to test them, and to further develop them.

It can be assumed that our mind shares a common plane with the material world that is not recognizable as such by humans. Thus, everything we believe we know and understand is shaped by the interaction of mind and matter. And it is precisely in this creative interaction that the basis for what we call the divine might lie.

There are a multitude of possible reasons why micro-PK is so difficult to prove. It is possible that researchers, through their personal scientific endeavors and the influence these exert on matter, are preventing the verification of their own assumptions. Perhaps our innate psi capacities are so difficult to detect precisely because their manifestation is completely unobtrusively embedded within the laws of nature.

A conclusive, less inconsistent body of evidence for these phenomena would shake our worldview to its core. Perhaps a natural, balancing

countermechanism is preventing definitive proof of these paranormal phenomena in order to maintain the natural balance of the world.

In psi research, we cannot currently expect any classical scientific proof of micro-PK. Rather, it is appropriate to gradually make psychokinetic processes tangible in order to analyze their occurrence and fluctuations and thus gain a better understanding of the mysterious processes in our universe.

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